

Lecture 18: Epilogue

Caltech CS156 Technical Study Guide

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Executive Summary

The final lecture synthesizes the entire Caltech CS156 curriculum into a unified map of machine learning theory, techniques, methods, and paradigms. It examines the Bayesian Learning framework and introduces Aggregation/Ensemble methods (Bagging, Boosting, and Stacking).

Master Map of Machine Learning Concepts (Lecture 18 Epilogue)



Figure 1: The Master Taxonomy of Machine Learning (Caltech CS156 Epilogue).

1 The Bayesian Learning Framework

Bayesian learning treats parameters $\mathbf{w}$ or hypotheses h as random variables with prior probability distributions $P(h)$.

1.1 Bayes' Rule for Hypotheses

$$P(h|\mathcal{D}) = \frac{P(\mathcal{D}|h)P(h)}{P(\mathcal{D})} = \frac{P(\mathcal{D}|h)P(h)}{\int_{\mathcal{H}} P(\mathcal{D}|h')P(h')dh'} \quad (1)$$

- $P(h)$: **Prior Distribution** representing subjective beliefs before seeing data.
- $P(\mathcal{D}|h)$: **Likelihood** of dataset $\mathcal{D}$ given hypothesis h .
- $P(h|\mathcal{D})$: **Posterior Distribution** updated after observing data.

1.2 Posterior Predictive Distribution

Rather than choosing a single hypothesis, Bayesian prediction integrates over the entire posterior space:

$$P(y|\mathbf{x}, \mathcal{D}) = \int_{\mathcal{H}} P(y|\mathbf{x}, h)P(h|\mathcal{D})dh \quad (2)$$

2 Aggregation Methods (Ensembles)

Aggregation combines multiple hypotheses $h_1, \dots, h_T$ into an ensemble output $H(\mathbf{x}) = \sum \alpha_t h_t(\mathbf{x})$.

2.1 Core Aggregation Strategies

1. **Bagging (Bootstrap Aggregating):** Trains hypotheses independently on T bootstrap samples drawn with replacement from D . Reduces variance $\text{Var}(g)$.
2. **Boosting (e.g., AdaBoost):** Trains hypotheses sequentially, reweighting misclassified examples from step $t - 1$ to force h_t to focus on hard errors. Reduces bias.
3. **Blending / Stacking:** Uses a meta-learner trained on out-of-fold validation predictions to determine optimal aggregation weights α_t .